

Cosmic Dodge

README / Project Documentation

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Project Type: Python & Pygame 2D Arcade Game

1. Project Overview

Cosmic Dodge is a 2D arcade-style space game developed using Python and Pygame. The player controls a character near the bottom of the screen and must avoid randomly falling stars. The objective is to survive for as long as possible while the game gradually becomes more challenging.

2. Features

- Player movement using the keyboard
- Randomly generated falling stars
- Collision detection
- Increasing difficulty
- Survival-time display
- Game-over screen
- Modular Python file structure
- Basic automated collision tests

3. Technologies Used

Python 3, Pygame, unittest, and Git/GitHub.

4. Project Structure

File	Purpose
main.py	Starts the game.
game.py	Main game controller and loop.
config.py	Game settings and constants.
player.py	Player movement and drawing.
star.py	Falling-star creation and management.
collision.py	Collision detection.
ui.py	HUD and game-over interface.
utils.py	Utility functions.
test_collision.py	Basic automated collision tests.

5. Installation

Install the required Python package:

```
pip install -r requirements.txt
```

6. How to Run

Run the game from the project folder:

```
python main.py
```

7. Controls

Left Arrow: Move left

Right Arrow: Move right

R: Restart after game over

Q: Quit

8. Game Objective

Avoid falling stars and survive for the longest possible time. The game becomes harder as time passes.

9. Testing

Run the basic collision tests with:

```
python -m unittest test_collision.py
```

10. GitHub Screenshot

Place output.png in the repository root and use this Markdown in README.md:

```
![Cosmic Dodge](output.png)
```

11. Author

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12. Educational Purpose

The project demonstrates Python concepts including functions, loops, conditions, lists, random number generation, keyboard input, timing, movement, collision detection, modular programming, and basic testing.